

For any technology

- ✓ Compute (CPU + Memory)
- ✓ Storage
- ✓ Network

Computer Network:

- ✓ Computer Network allows the computers to talk to each, for data and resource sharing.
- ✓ We need an **IP address** for every computer.
 - IPv4 address
 - Old way to assign the IP addresses.
 - IPv6 address
 - New way to assign Ips
 - This got introduced due to the shortage of IPv4 address.
- ✓ Types of addresses:
 - Public IP address
 - Are the Ips that are used over the internet.
 - PIP is allocated via ISP (Internet Service Provider) → Airtel, BSNL, Jio, etc.
 - Private IP address
 - These are used within home, organization, building.
 - Range:
 - Class A → 10.X.X.X
 - Class B → 172.16.X.X
 - Class C → 192.168.X.X
 - Private Ips can be configured by us or ISP will configure.

Server

- ✓ A server always “serves”.
- ✓ A server could be
 - A physical or a virtual machine.
 - A windows operating system or a linux operating system.

Client

- ✓ Always seeks for a request from a server.
 - A physical or a virtual machine.
 - A windows operating system or a linux operating system.

Windows Operating System		Linux Operating System	
Server OS	Client OS	Server OS	Client OS
Windows Server 2003	Window XP	RHEL 6/7/8/9/10	Ubuntu
Window server 2008	Window Vista	CentOS 6/7/8/9/10	Garuda Linux
Window server 2008R2	Window 7	Suse linux	Kali linux (Pentesting)
Window server 2012	Window 8.0	Fedora	Back Track (Pentesting)
Window server 2012R2	Window 8.1	Solaris (UNIX)	Parrot linux (Pentesting)
Window server 2016	Window 10	HPUX (UNIX)	
Window server 2019	Window 11		
Window server 2022			
Window server 2025			

Storage:

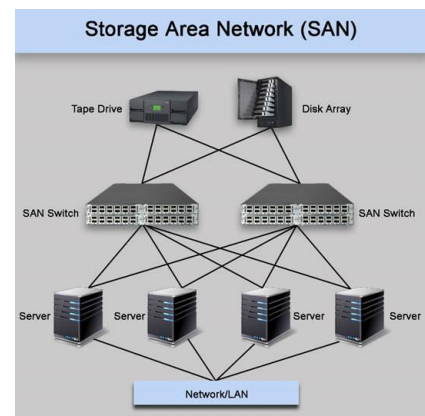
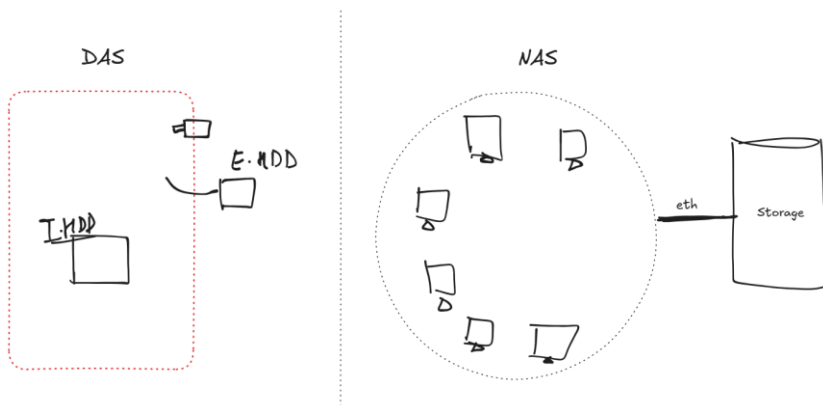
✓ 3 types of storage:

○ DAS

- Direct Attached Storage.
- This type of storages is directly connected to the server or via small cable.
- Ex:
 - Pen drive
 - External HDD
 - Internal HDD
- This is a faster way to store the data.
- But no redundancy.

○ NAS

- Network Attached Storage.
- Whole network is connected to this storage within a LAN through ethernet cable.
- No redundancy.
- Easy to access and easy to share.



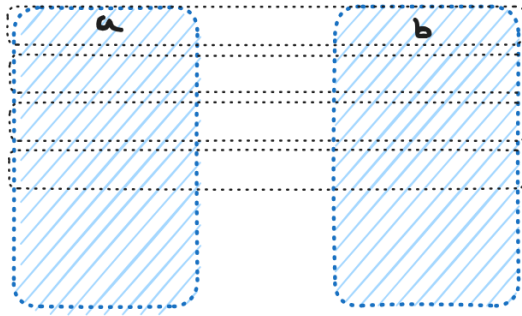
SAN storage

- ✓ Storage Area Network
- ✓ It is a production-level solution because it offers/provides redundancy.

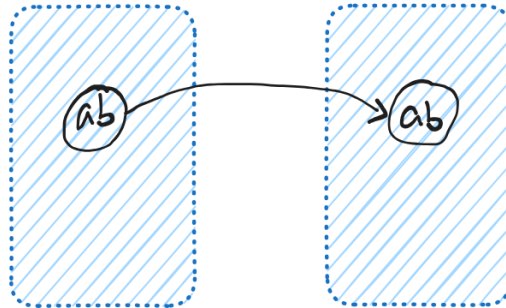
Storage Disks

- *Basic disks*
 - These types of disks cannot be expanded.
 - It remains as a fixed disk.
- *Dynamic disks*
 - These types of disks are expandable (increase or decrease the size of the disk).
 - Dynamic disks are of 3 types:
 - Stripe volume → RAID 0
 - In this minimum of 2 identical disks are required.
 - In this, logical stripes are created on the disks to store data in faster manner.
 - Writing on disk is faster, reading is slower.
 - Mirror volume
 - In this also, minimum of 2 identical disks is required.
 - Data gets stored in one disk, and immediately a copy of data gets stored in 2nd disk.
 - Spanned volume
 - Here 2 or more disks of identical or unidentical sizes are required.
 - Here, we logically club disks together to make one logical partition out of it.

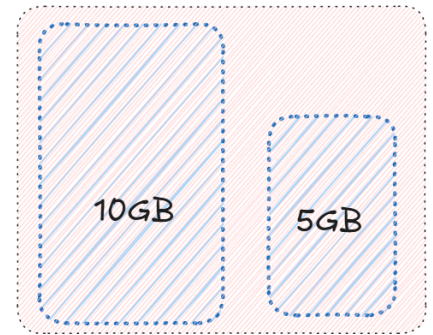
Stripe volume



Mirror volume



~15GB



Spanned volume

- Basic disk → Dynamic disk → data lose
- Dynamic disk → Basic disk → data lose

Workgroup

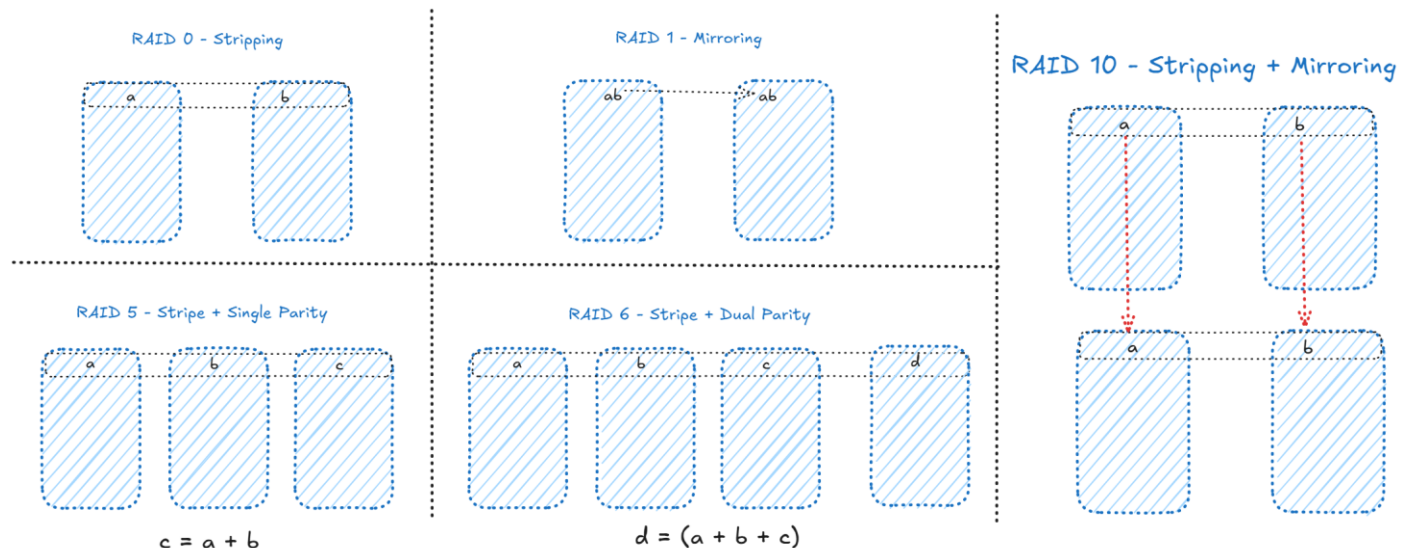
- In windows environment, a standalone machine that is **not centrally managed** but managed using individual administrator is considered as a workgroup machine.
- Home laptops, desktop uses workgroup environment.

Domain

- A domain machine is a machine that is connected and managed centrally using a domain controller (DC).
- Generally, organizations use this domain environment to manage the servers and client devices centrally.

RAID

- ✓ Redundant Array of Independent Disk
- ✓ Different RAID concepts:
 - RAID 0 → Stripe
 - RAID 1 → Mirror
 - RAID 5 → Stripping with single parity
 - RAID 6 → Stripping with dual parity
 - RAID 10 → Stripping with Mirroring



File systems in windows:

- ✓ FAT (File Allocation Table) or FAT16
- ✓ FAT32
- ✓ **NTFS** (New Technology File System) → default
- ✓ ReFS (Resilient File System) → Mirroring (RAID 1)

File systems in Linux:

- ✓ EXT (**E**xtended File System)
- ✓ EXT2
- ✓ EXT3
- ✓ EXT4
- ✓ XFS (**e**Xtended File System → default)
- ✓ BTRFS
- ✓ ExFAT

Windows Server Administration

Window Server 2016/2019/2022/2025 Editions:

- ✓ **Essential**
 - Up to 25 users and 50 devices are allowed.
- ✓ **Standard**
 - Up to 2 VMs.
- ✓ **Data Center**
 - Unlimited virtual machines

Role and Features:

- ✓ Role or feature or both are required in order to work on a windows server.
- ✓ **Role**
 - A role is a functionality that a server must perform.
 - A role can be installed using:
 - Server Manager Dashboard Page → GUI
 - PowerShell → CLI
 - Example:
 - DHCP - Dynamic Host Configuration Protocol
 - It provides dynamic IP address
 - DNS - Domain Name System
 - It resolves Name into IP address
 - IIS web server - Internet Information Service
 - This hosts a website on windows
- ✓ **Features**
 - They are the supporting tools for any role.
 - When a role is installed, features also gets installed in the backend.
 - You can install additional features manually.
 - Example:
 - NLB (Network Load Balancer)
 - FC (Failover Cluster)
 - DOT NET Framework (4.8)

Agenda

- Install a windows server 2016/2019 operating system on VMWare Workstation
- Perform post-installation configuration → Server Dashboard Page:
 - Stop firewall
 - Change the IP address, Subnet mask, default gateway and DNS.
 - Change the time zone
 - IE Enhanced security configuration → Off.
 - Change the computer name → restart the PC.
- A total of 3 VMs are needed
 - 1st VM → Domain Controller
 - 2nd VM → Node01
 - 3rd VM → Node02

To install, we can do it using:

1. GUI → **Desktop Experience**
2. CLI → **Server Core OS**

Note – whenever you install a windows server operating system, by default username “Administrator” will be created. You can only set the password for this Administrator → pass@word1

Storage Spaces:

- ✓ It refers to adding of multiple disks and make one-logical disk out of it.
- ✓ We will add 2 disks (each of 10 GB) to the DC.

Partition style

1. MBR
 - a. Master Boot Record
 - b. It supports maximum 4 primary partitions.
 - c. It supports max of 2 TB of HDD.
2. GPT
 - a. GUID (Globally Unique Identifier) Partition Table.
 - b. It supports maximum 128 primary partitions.
 - c. It supports max of 18 EB (18×10^6 GB) of HDD.